

SIOP Reporting – Power BI Solution

Case Study 2 Submission Report

Internship Assessment Submission

Project Overview

The objective of this case study was to build an interactive Power BI solution that enables users to evaluate forecast performance by comparing multiple forecast snapshots against actual outcomes.

The source data contained different versions of forecasts generated over time. To support meaningful analysis, these snapshots were transformed into business-friendly lag views:

- Resultant (latest forecast)
- Lag 0 (previous forecast)
- Lag 1 (oldest forecast)

The final dashboard was designed to answer practical business questions such as:

- How accurate are our forecasts?
 - Are we consistently over-forecasting or under-forecasting?
 - Which countries contribute most to forecast volumes?
 - Which regions require further investigation?
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Dataset Summary

The solution was developed using two source datasets:

Forecast Dataset

- 246,477 rows
- 15 columns
- 3 forecast snapshots:
 - 2023 P10
 - 2023 P11
 - 2023 P12

Actual Dataset

- 500,000 rows

- 6 columns
- Item-Country-Period level actual quantities

During profiling, a significant data quality issue was identified. Approximately 88% of the Actual dataset consisted of duplicate records, requiring extensive cleansing before analysis.

Data Preparation

Several preprocessing steps were performed before modelling:

- Removed 6,129 duplicate Forecast records.
- Removed 442,054 duplicate Actual records.
- Aggregated remaining duplicate business keys.
- Standardised mixed Item identifiers into text format.
- Converted forecast snapshots into Resultant, Lag 0, and Lag 1 structures.
- Merged Actual and Forecast data using Period, Item, and Country.

One important decision was constructing the final dataset using the union of all snapshot keys rather than relying only on the latest forecast. This ensured that earlier periods available in Lag 1 were not lost during transformation.

Data Model

The final model consisted of:

Fact_SIOP

Containing:

- Forecast quantities
- Actual quantities
- Business keys

Dimension Tables

- Dim_Period
- Dim_Country
- Dim_LagSelector

The Lag Selector was intentionally left disconnected to support dynamic DAX calculations without directly filtering the fact table.

Measures Implemented

The dashboard included several analytical measures:

- Total Forecast Quantity
- Total Actual Quantity
- Forecast Accuracy %
- Forecast Bias %
- Forecast Variance
- Absolute Error Quantity

Forecast Accuracy and Bias measures were designed to return blank values whenever actual quantities were unavailable, preventing misleading results.

Dashboard Overview

The final dashboard focused on providing a concise executive view of forecast performance.

Key components included:

- Average Forecast Accuracy
- Forecast Variance
- Absolute Error Quantity
- Periods with Actuals
- Total Forecast Quantity by Country
- Actual vs Forecast trends across periods
- Forecast Accuracy gauge
- Absolute Error distribution by Country

This design enables stakeholders to identify performance gaps quickly while maintaining a simple and intuitive user experience.

Validation and Testing

A series of validation checks were performed to ensure the reliability of the solution.

Key validations included:

- Forecast records reduced from 246,477 to 240,348 after duplicate removal.

- Actual records reduced from 500,000 to 53,276 after cleansing and aggregation.
- Forecast measures switched correctly between Resultant, Lag 0, and Lag 1 selections.
- Future periods returned blank accuracy values instead of artificial percentages.
- Dashboard filters updated visuals consistently.
- Fact table actual totals matched staging totals exactly, confirming that joins did not introduce row inflation.

These checks increased confidence in both the transformation process and the analytical outputs.

Key Findings

The completed dashboard indicated that forecast performance was moderate, with an average forecast accuracy of approximately 54%.

The analysis also revealed noticeable forecast variance across countries, suggesting that forecasting quality differed by region and that some markets may require additional review.

The ability to compare forecast versions provided useful visibility into how forecast assumptions evolved over time.

Conclusion

This case study demonstrated the importance of combining data preparation, validation, and business-focused reporting to produce reliable analytical solutions.

Beyond building dashboards, the project reinforced the value of understanding source data, validating transformations carefully, and presenting insights in a way that supports decision-making.

The final Power BI solution provides an interactive and practical approach to monitoring SIOP forecast performance and identifying opportunities for continuous improvement.